

35. Scatter Plot Comparing Two Subject Marks

Aim

To write a Python program to draw a scatter plot comparing marks obtained in two subjects.

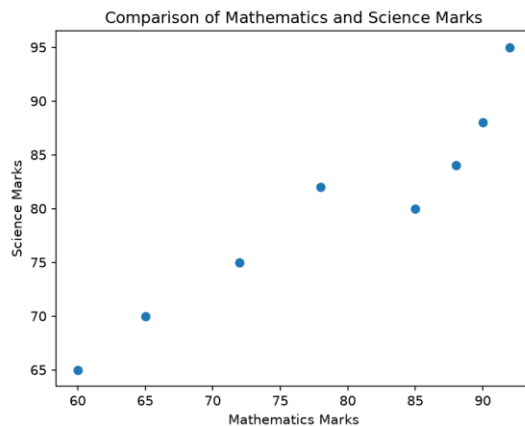
Dataset

Student	Mathematics	Science
Anu	85	80
Bala	72	75
Cathy	90	88
David	65	70
Esha	78	82
Farhan	92	95
Gita	60	65
Hari	88	84

Python code:

```
1 import matplotlib.pyplot as plt
2 maths = [85, 72, 90, 65, 78, 92, 60, 88]
3 science = [80, 75, 88, 70, 82, 95, 65, 84]
4 plt.scatter(maths, science)
5 plt.xlabel("Mathematics Marks")
6 plt.ylabel("Science Marks")
7 plt.title("Comparison of Mathematics and Science Marks")
8 plt.show()
```

Output



Result:

The relationship between Mathematics and Science marks was successfully represented using a scatter plot.

Dataset

36. Scatter Plot for Three Groups Comparing Weights and Heights

Aim

To write a Python program to draw a scatter plot for three different groups comparing their weights and heights.

Dataset

Group	Height (cm)	Weight (kg)
Group A	150	50
Group A	155	53
Group A	160	55
Group B	165	60

Group	Height (cm)	Weight (kg)
Group B	170	65
Group B	175	68
Group C	180	72
Group C	185	78
Group C	190	82

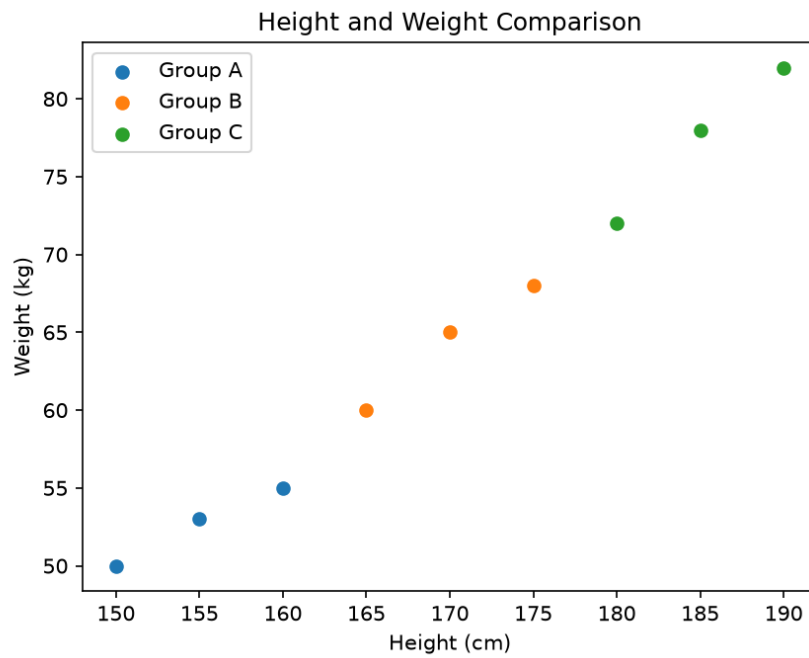
Python code:

```

1  import matplotlib.pyplot as plt
2  height_A = [150, 155, 160]
3  weight_A = [50, 53, 55]
4  height_B = [165, 170, 175]
5  weight_B = [60, 65, 68]
6  height_C = [180, 185, 190]
7  weight_C = [72, 78, 82]
8  plt.scatter(height_A, weight_A, label="Group A")
9  plt.scatter(height_B, weight_B, label="Group B")
10 plt.scatter(height_C, weight_C, label="Group C")
11 plt.xlabel("Height (cm)")
12 plt.ylabel("Weight (kg)")
13 plt.title("Height and Weight Comparison")
14 plt.legend()
15 plt.show()

```

Output



Result:

The heights and weights of three groups were successfully compared using a scatter plot.